

MAX HALANEN

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TECHNICAL SKILLS

Languages: C/C++, Python, Verilog, Assembly (x86, ARM, RISC-V), Powershell, Bash, Java

Tools: Git, GDB, QEMU, Valgrind, FreeRTOS, Docker, KiCad, Linux, Doxygen, Jira (Agile), Neovim, Eclipse

Networking: Wireshark, Lightweight IP (lwIP), TCP/IP, ESXi, vSphere, PuTTY

Embedded & Hardware: ARM TrustZone, STM32, ESP32, Oscilloscope, Logic Analyzer, USART, SPI, I2C

EXPERIENCE

General Dynamics May 2026 - Present
Software Developer - Internship Ottawa, ON

Carleton University Jan 2026 - Present
Embedded Security Researcher (Part-Time) Ottawa, ON

- Implemented **secure device attestation** on ESP32-C6 using ECDSA-signed tokens for verification
- Analyzed isolation differences between ESP-TEE and ARM TrustZone, **identifying key security tradeoffs**
- **Built a secure device-server protocol** to remotely validate trusted hardware through cryptographic attestation

General Dynamics Sep 2025 - Dec 2025
Embedded Software Developer - Internship Ottawa, ON

- Led software migration to a secure ARMv8 processor, **reducing product lifecycle costs by \$125k**
- **Abstracted I2C interface to GPIO**, enabling a standard CMSIS GPIO interface for an I/O expander
- Created containerized dev and build environments to **streamline CI/CD** and ensure consistent development
- Developed a WS2812 LED driver using **DMA-driven PWM** to eliminate CPU load
- Implemented FreeRTOS CLI accessible via TCP and UART enabling runtime debug and statistics

General Dynamics Sep 2024 - Aug 2025
Systems Engineer - Internship Ottawa, ON

- Developed a router configuration automation app, **saving \$25,000** in projected salary costs
- Created remote installer deploying software across **100+ machines** using PowerShell and WinRM
- Worked on a Wireshark packet dissector, opening up new capabilities in debugging and automation

PROJECTS

Handheld GPS device | C/C++, ESP32, GPS

- Streamed maps from SD card to TFT LCD via SPI using a graphical library (LVGL) to render map tiles
- Integrated GPS data and implemented coordinate-to-tile algorithm to render correct map tiles in real time

IoT Thermostat System | C/C++, STM32, lwIP, FreeRTOS

- Built an IoT thermostat with FreeRTOS + lwIP and BMP180 sensor, achieving **99%** idle task uptime
- Implemented UDP server to transmit sensor data to home HTTP server for real-time monitoring

RISC-V Processor | Verilog, CPU Design, RISC-V

- Built the initial stages of a RISC-V CPU in Verilog, including fetch/decode, immediate logic, and control FSM

EDUCATION

Carleton University Graduating: **May 2027** | GPA: 3.9 / 4.0
Bachelor of Computer Science Honors Ottawa, ON

Relevant Courses: Discrete Mathematics and Algorithms II (A+), Database Management Systems (A+), Introduction to Systems Programming (A), Applied Cryptography and Authentication (A+), Operating Systems (A+)